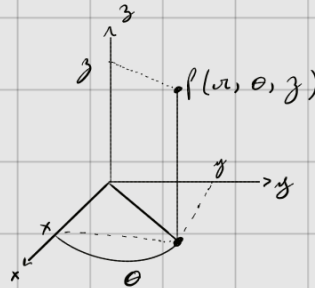


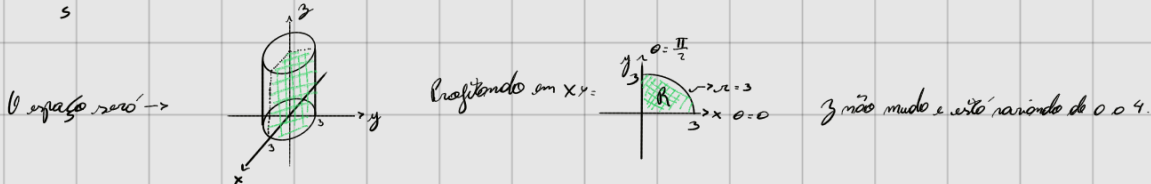
Coordenadas Cilíndricas:

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases} \quad \text{O valor de } r \text{ também é igual a } \rho.$$



$x = r \cos \theta$	$x^2 + y^2 = r^2$
$y = r \sin \theta$	$dV = r \, dr \, d\theta \, dz$
$z = z$	

Calcular $\iiint_S \sqrt{x^2 + y^2} \, dV$ sendo S determinado pelas superfícies $z=0$, $z=4$ e $x^2 + y^2 = 9$ no 1º octante.

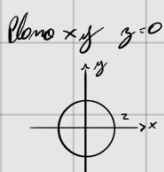


$$R = \begin{cases} 0 \leq z \leq 4 \\ 0 \leq r \leq 3 \\ 0 \leq \theta \leq \frac{\pi}{2} \end{cases} \quad f(x, y, z) = \sqrt{x^2 + y^2} \rightarrow f(r, \theta, z) = \sqrt{r^2} = r, \quad r \geq 0$$

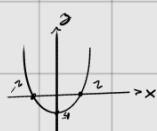
$$\text{A integral fica } \int_0^{\pi/2} \int_0^3 \int_0^4 r \cdot r \, dz \, dr \, d\theta = \int_0^{\pi/2} \int_0^3 r^2 \cdot 4 \, dr \, d\theta = 4 \cdot \int_0^{\pi/2} \left[\frac{r^3}{3} \right]_0^3 d\theta = 4 \cdot \int_0^{\pi/2} \left(\frac{27}{3} - 0 \right) d\theta =$$

Exercício Calcular $\iiint_{R^2} \sqrt{x^2 + y^2} \, dV$, a região delimitada pelas superfícies $z = x^2 + y^2 - 4$ e $z = 4 - x^2 - y^2$

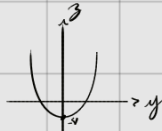
Ⓘ $z = x^2 + y^2 - 4$



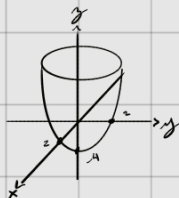
Plano xz y=0



Plano yz x=0



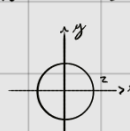
$$z = x^2 - 4$$



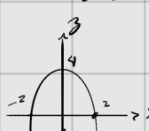
* Parabolóide
para cima

Ⓡ $z = 4 - x^2 - y^2$

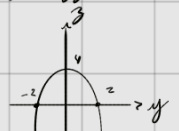
Plano xy z=0



Plano xz y=0

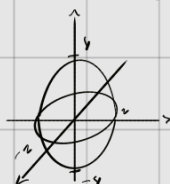


Plano yz x=0



* Parabolóide
para baixo

Resultado:



$$r^2 = 4 \rightarrow r = 2$$

$$\begin{cases} x^2 + y^2 - 4 \leq z \leq 4 - x^2 - y^2 \\ 0 \leq r \leq 2 \\ 0 \leq \theta \leq 2\pi \end{cases}$$

$$\int_0^{2\pi} \int_0^2 \int_{x^2+y^2-4}^{4-x^2-y^2} \sqrt{x^2+y^2} \cdot r \, dz \, dr \, d\theta = \int_0^{2\pi} \int_0^2 r^2 (4 - r^2 - (r^2 - 4)) \, dr \, d\theta = \int_0^{2\pi} \int_0^2 8r^2 - 2r^4 \, dr \, d\theta =$$

$$\int_0^{2\pi} \left[\frac{8r^3}{3} - \frac{2r^5}{5} \right]_0^2 d\theta = \int_0^{2\pi} \left(\frac{128}{3} - \frac{64}{5} \right) d\theta = \frac{128}{15} \cdot 2\pi = \frac{256}{15} \pi //$$

$$\frac{64}{3} - \frac{64}{5} = \frac{320 - 192}{15} = \frac{128}{15}$$